Ziyan SONG

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EDUCATION

Brown University **Sep 2024 – May 2026**
Master of Science in Computer Science, GPA: 4.0/4.0 *Providence, RI*

Northeastern University **Sep 2020 – Jun 2024**
Bachelor of Engineering in Software Engineering, GPA: 3.9/4.0 *Liaoning, CN*

WORK EXPERIENCE

Aigeo Inc. **Nov 2025 – Feb 2026**
Founding Software Engineer *San Francisco, CA*

- Architected and launched an AI search visibility monitoring and optimization platform using React and Node.js, reaching $50K+ ARR and 20+ B2B clients within 4 months.
- Engineered a high-concurrency pipeline orchestrating 200+ parallel LLM queries across 7 providers using PostgreSQL-backed job queue and async workers with built-in retry logic for fault tolerance.
- Built core SaaS infrastructure including RBAC, OAuth 2.0, and Stripe subscription billing, with real-time analytics dashboards aggregating GA4 and SerpApi for cross-source visibility insights.
- Containerized and deployed auto-scaling services with Docker and AWS, establishing automated CI/CD pipelines via GitHub Actions supporting rapid daily deployment.

Fanchatsy Inc. **May 2025 – Aug 2025**
Software Engineering Intern *Boston, MA*

- Delivered a personalized push notification system for iOS using APNs and Python APScheduler, tracking user activity and delivering behavior-triggered notifications to 1,000+ users.
- Optimized AI chat context by merging recent messages with long-term memory and dynamically generating summaries, reducing LLM token usage by 35% while maintaining coherent multi-turn conversations.
- Built a subscription system using StoreKit 2 and FastAPI, handling receipt validation and Apple Server-to-Server notifications to ensure consistent state synchronization across payment lifecycle events.

Bosch Automotive Products CO., Ltd. **Feb 2023 – Aug 2023**
Software Engineering Intern *Jiangsu, CN*

- Implemented high-performance IPC using shared memory and refactored middleware APIs to support dynamic configuration, improving system flexibility across heterogeneous environments.
- Designed and developed a time management module for multi-threaded system simulations, reducing function call latency by 18% and achieving 100% test coverage via GTest.
- Automated cross-compilation and benchmarking pipelines using Python and Shell scripts, improving testing efficiency by 90% and enabling end-to-end report generation in under 3 minutes.

PROJECT EXPERIENCE

LLM-powered Streaming Data Processing System | *Course Project* **Feb 2025**

- Designed and implemented an LLM-integrated streaming data processing system, enabling an asynchronous push-based execution model using Python's asyncio module.
- Developed multiple semantic operators, including filtering, aggregation and successive event identification, to optimize traditional rule-based Complex Event Processing(CEP), achieving an F1-score of 0.93.
- Engineered time-based, index-based and semantic-based windowing mechanisms for automated data segmentation.
- Unified LLM invocation interfaces across diverse backends, supporting OpenAI API, local Ollama models and VLLM.

Relational Database Engine | *Course Project* **Oct 2024**

- Engineered a high-performance RDBMS engine in Go, implementing B+ tree index, LRU-based buffer pool, and a disk manager to support scalable data storage and concurrent access.
- Developed a transaction manager with Strict 2PL and ARIES recovery, ensuring ACID compliance and reliability under high-throughput workloads.
- Optimized query execution with a Bloom-filter-accelerated hash table and Grace Hash Join, achieving a 5× speedup in join performance.

TECHNICAL SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, C++, Golang
Frameworks & Tools: React, Next.js, Tailwind CSS, Spring Boot, Prisma, Node.js, Astro, Jenkins, Docker, Kubernetes
Databases & Cloud : MySQL, PostgreSQL, MongoDB, Redis, Kafka, AWS, GCP